Usama Bashir Butt

Full-Stack AI Engineer

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Professional Summary

Full-Stack AI Engineer with 5 years of experience developing production applications, specializing in AI/LLM integrations, RAG systems, AI agents, and backend engineering. Experienced with Python, FastAPI, ASP.NET Core, Angular, React, PostgreSQL, and Docker.

Work Experience

August-Wilhelm Scheer Institut gGmbH Saarbrücken, Germany
Full Stack Developer August 2023 – Present

- Built production LLM applications with Angular, ASP.NET Core, PostgreSQL, Docker and S3-compatible storage, integrating multiple LLM APIs (OpenAI, Anthropic) to deliver context-aware conversational features.
- Designed and deployed RAG pipelines using Python, FastAPI, and LangChain, combining Neo4j knowledge-graph retrieval with hybrid keyword/semantic search powered by Sentence Transformers, pgvector, and HNSW indexing.
- Developed multi-agent AI workflows with LangGraph, using tool calling with Pydantic-validated schemas and iterative retrieval to support multi-step reasoning and automated information retrieval.
- Delivered LLM-based workflow automation for 5+ business processes using REST APIs, webhooks and the OpenAI and Anthropic APIs, reducing manual support effort by 20%.
- Built LLM evaluation and monitoring workflows using LLM-as-a-judge and user feedback techniques to measure answer relevance, reliability and consistency, and tracked token usage, latency and cost per response to detect quality regressions.
- Implemented guardrails for predictable AI behavior in production: Pydantic schema validation of structured LLM outputs, input validation, and secure secret management with HashiCorp Vault and envelope encryption for keys and database contents.
- Built and documented REST APIs and service layers in Python (FastAPI) and C# (ASP.NET Core), applying repository patterns, DTOs and Swagger documentation to integrate AI capabilities cleanly into existing products.
- Containerized services using Docker and Docker Compose, configured Nginx, and automated deployments through Azure DevOps CI/CD to reduce manual deployment effort.
- Collaborated with product owners and non-technical stakeholders to translate business requirements into AI solution designs, communicating technical trade-offs and delivering features from prototype to production.
- Delivered frontend and backend features across 10+ applications (Angular, React, TypeScript, C#, ASP.NET Core) and built four reusable Storybook/Nx UI libraries adopted by 3 teams, reducing cross-team code duplication by 40%.
- Wrote 50+ unit, integration and end-to-end tests using Jest and Cypress to validate functionality.

Hysab Kytab Pvt. Ltd. Karachi, Pakistan
Associate iOS Developer August 2021 – May 2023

- Delivered 5 native iOS and web features for an SDK integrated into a banking application serving 1M+ users, from requirements analysis through development, testing, and production release.
- Integrated Hysab Kytab SDK and built a WKWebView JavaScript bridge to enable secure authentication and communication between native iOS and web components.
- Refactored legacy iOS modules using MVVM architecture and reusable services, reducing application startup time by 70% (from 10 to 3 seconds) while improving maintainability.

Education

Universität des Saarlandes Saarbrücken, Germany
Master of Science (M.Sc.) in Data Science & Artificial Intelligence Expected 2027

National University of Computer and Emerging Sciences (NUCES – FAST) Karachi, Pakistan
Bachelor of Science (B.Sc.) in Computer Science 2021

Skills

Programming Languages: Python, C#, TypeScript, JavaScript (ES6+), SQL
Frontend: Angular, React, HTML5, CSS3, RxJS, NgRx, Storybook, WCAG, Responsive Design
Backend: ASP.NET Core, FastAPI, Node.js, REST APIs, NuGet, GraphQL, WebSockets, Microservices
AI & LLM: RAG, LLM APIs, AI Agents, MCP, LangChain, LangGraph, Hugging Face Transformers, n8n, Claude Code, Codex
Databases: PostgreSQL, pgvector, Neo4j, MySQL, SQL Server, Entity Framework Core
DevOps & Testing: Docker, Docker Compose, Azure DevOps, CI/CD, Nginx, AWS, Kubernetes, Git, Jest, Cypress
Security: JWT, HashiCorp Vault, Envelope Encryption, Keycloak
Languages: English (C1), German